

Homework 7.2

Problem 1

A)

$$S \rightarrow A \mid D$$

$$A \rightarrow aAa \mid bAb \mid B$$

$$B \rightarrow aCb \mid bCa$$

$$C \rightarrow aC \mid bC \mid \lambda$$

$$D \rightarrow EF$$

$$E \rightarrow aEb \mid \lambda$$

$$F \rightarrow aF \mid \lambda$$

B)

$$S \rightarrow AD$$

$$A \rightarrow aAa \mid bAb \mid B$$

$$B \rightarrow aCb \mid bCa$$

$$C \rightarrow aC \mid bC \mid \lambda$$

$$D \rightarrow EF$$

$$E \rightarrow aEb \mid \lambda$$

$$F \rightarrow aF \mid \lambda$$

C)

$$S \rightarrow ASD \mid \lambda$$

$$A \rightarrow aAa \mid bAb \mid B$$

$$B \rightarrow aCb \mid bCa$$

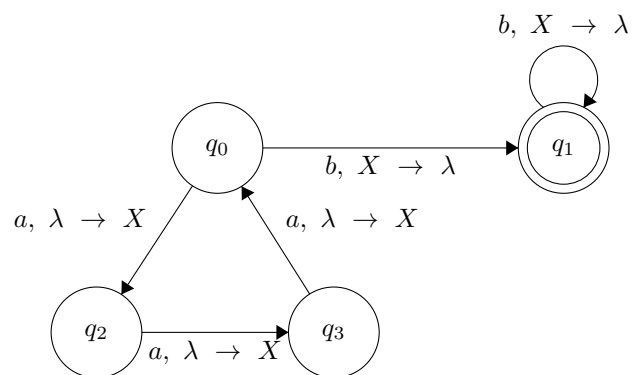
$$C \rightarrow aC \mid bC \mid \lambda$$

$$D \rightarrow EF$$

$$E \rightarrow aEb \mid \lambda$$

$$F \rightarrow aF \mid \lambda$$

Problem 2



Problem 3

No, removing strings can cause a language to no longer be deterministic